

LLM-Based Crypto Market Signal Detection from X



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Motivation

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- **Market Vulnerability:** While cryptocurrencies are popular investments, the markets are actually highly fragile and volatile
- **Social Media Impact:** Public figures and influencers possess considerable power over market trends, where a single social media post can trigger financial shifts
- **Project Objective:** The goal is to apply Large Language Models (LLMs) to filter influential posts and evaluate their exact impact on the market.

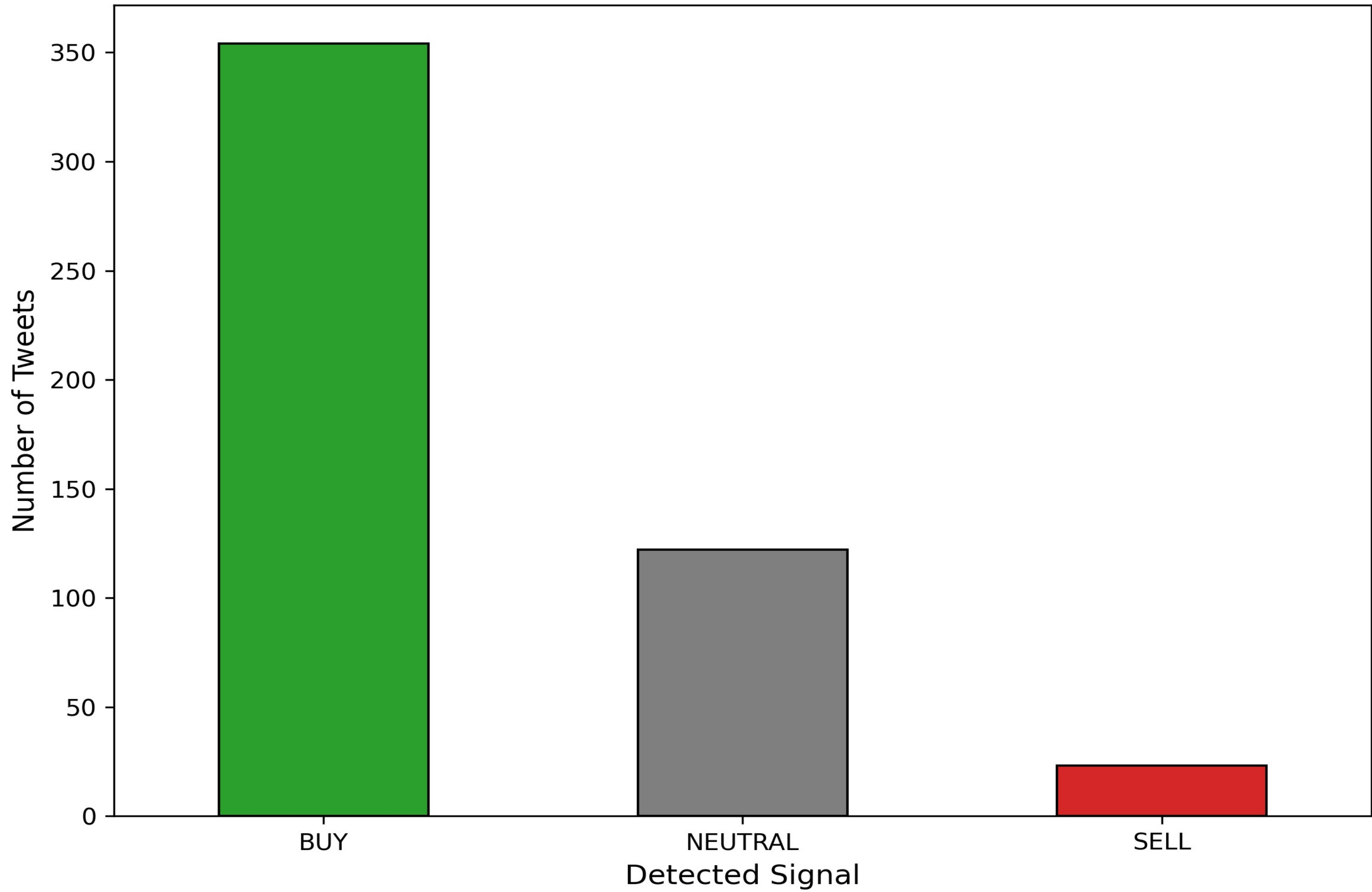
Other Motivation

- **Personal Experience:** Experiencing firsthand how influencer hype (like Elon Musk and Dogecoin) directly drives individual investment decisions.

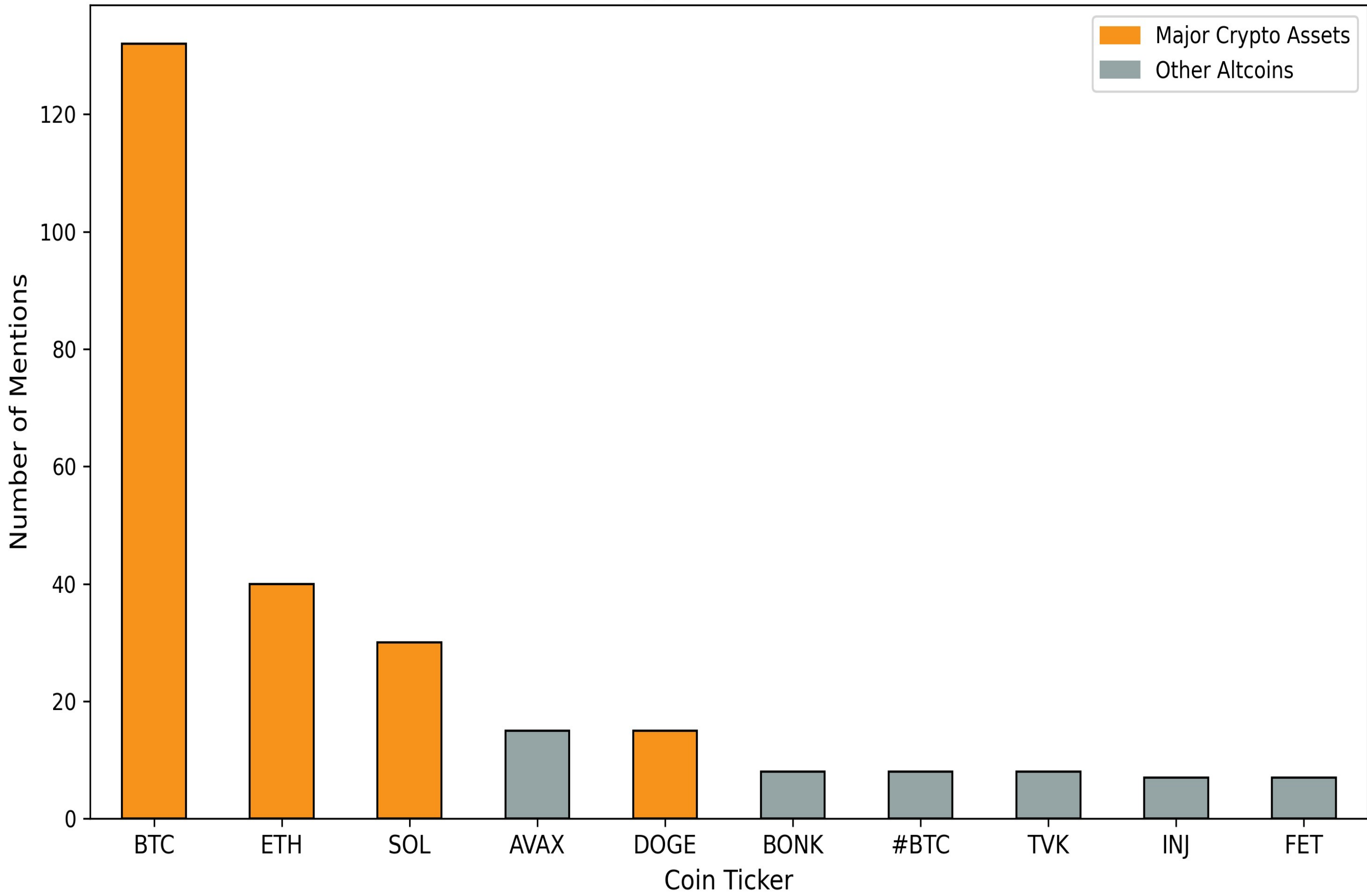
Implementation

- **Data Source:** We automatically get a list of recent tweets about cryptocurrencies from online dataset sources
- **AI Analysis (Llama 3.1):** We send the text of each tweet to a powerful AI model that reads it just like a human would
- **Result:** The AI classifies the sentiment: Is it positive ("BUY") , neutral ("NEUTRAL") or negative ("SELL")?
- **Detection:** Identifying posts that could potentially trigger market movements (e.g. Like/Retweet counts)
- **Confirmation:** Evaluating historical price data to verify whether a specific post actually caused a measurable impact on the cryptocurrency market.

Distribution of AI Signals (500 Crypto Tweets)



Most Mentioned Cryptocurrencies in 500 Tweets



Future Work

- **Scale Up:** The next phase will expand the processing pipeline from the initial 500-tweet sample
- **Price Correlation:** Future work will focus on overlaying these generated AI sentiment signals with actual historical cryptocurrency price charts to evaluate the true predictive power of social media trends

Resources

[ABC20] Stephan Akkerman. "Financial Tweets Dataset." Hugging Face, 2023. <https://huggingface.co/datasets/StephanAkkerman/financial-tweets>

[ABC21] Mendeley Data. "Cryptocurrency Sentiment Analysis Dataset." Mendeley Data, 2020. <https://data.mendeley.com/datasets/x7yvshrnxy/1>